

FactumDB

project proposal

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EXECUTIVE SUMMARY

MySQL investigations may involve separate InnoDB .ibd files and binary logs that must be manually correlated to understand how records changed over time. This project proposes a forensically sound tool that combines schema extraction, page-integrity checking, row unloading, and binary-log decoding to reconstruct MySQL record and transaction histories. Its main contribution is a correlation engine that links logged row events to physical records, groups events into transactions, detects missing or conflicting evidence, and compares log-derived states with the physical tablespace. Source evidence will be handled read-only, hashed with SHA-256, and analysed using verified working copies, while commands, tool versions, and provenance will be recorded for repeatability. The tool will provide readable timelines, reconciliation views, and reproducible reports, and will be evaluated using controlled synthetic datasets containing known transactions, evidence gaps, corruption, and unsupported inputs.

PROBLEM STATEMENT AND MOTIVATION

MySQL investigations may involve InnoDB .ibd files and binary logs that provide different views of database activity. An .ibd file may contain current or recoverable table records, while binary logs record available inserts, updates, and deletes. Utilities such as ibd2sdi, innochecksum, ibd2sql, and mysqlbinlog can extract schema information, validate pages, unload rows, and decode log events, but their outputs must still be interpreted and correlated manually.

Investigators may need to determine which events affected a specific record, which events belong to the same transaction, how a record changed over time, whether deleted records are supported by physical remnants, and whether the log-derived final state matches the tablespace. This becomes difficult when evidence contains many records, missing or rotated logs, partial row images, or inconsistent timestamps.

The proposed tool will integrate these outputs into traceable record histories and transaction timelines which is a gap in currently available MySQL database forensic tools. It will preserve source integrity, correlate row events with physical records, reconstruct available states, identify gaps or conflicts, and generate readable reports with provenance. The tool can support forensic investigators, incident-response teams, database administrators, and auditors, but it will not independently identify the responsible person or determine intent.

OBJECTIVE AND SCOPE

The main objective of this project is to design and implement a forensically sound tool that correlates records extracted from MySQL InnoDB tablespace files with row-based binary-log events in order to reconstruct record histories, group related changes into transactions, compare log-derived states with physical database records, and present the findings in an investigator-readable form.

The specific objectives are to:

- preserve source-evidence integrity through read-only handling, SHA-256 hashing, and verified working copies.
- extract and normalize schema, page-integrity, logical-row, and binary-log information using ibd2sdi, innochecksum, ibd2sql, and mysqlbinlog.
- map positional binlog columns to their corresponding table columns using schema metadata.
- group individual row events into committed transactions.
- correlate insert, update, and delete events with logical records using primary or unique keys.
- compare the expected state with the physical state extracted from the .ibd file;
- identify missing logs, incomplete histories, unsupported inputs, ambiguous matches, and conflicting evidence.
- provide transaction timelines, record-history views, reconciliation results, provenance details, and reproducible report exports.

The initial implementation will support one fixed MySQL 8.4.x release, the InnoDB storage engine, file-per-table .ibd files, row-based binary logs, full row images where available, explicit primary keys, unencrypted evidence, and a selected set of commonly used data types. The tool will perform offline analysis of previously acquired evidence files and will not connect to or modify a live MySQL server.

The first version will not support live acquisition, encryption bypass, redo- or undo-log parsing, statement-based log reconstruction, all MySQL versions, complete schema-evolution handling, guaranteed deleted-row recovery, or automatic attribution of database changes to a specific person. It will also not attempt to reconstruct the exact original SQL statements or make legal conclusions about intent or responsibility.

The expected deliverables are a working investigator-facing application, a documented correlation and reconstruction engine, synthetic test datasets, evaluation results, reproducible JSON, CSV, HTML, or PDF reports, source code, build and run instructions, a README, and a short user guide.

HIGH LEVEL APPROACH AND IMPLEMENTATION

Figure 1: High-Level Evidence Processing Pipeline

This flowchart illustrates the end-to-end processing stages of the forensic tool, demonstrating how original evidence is safely hashed, extracted using validated external utilities on read-only working copies, and normalized before being passed to the correlation engine.

Figure 2: Correlation Engine Architecture

This flowchart details the internal logic of the correlation engine, showing how normalized data is mapped to schemas, grouped into transactions, reconstructed chronologically, and reconciled against physical records to produce rule-based, reproducible correlation classifications

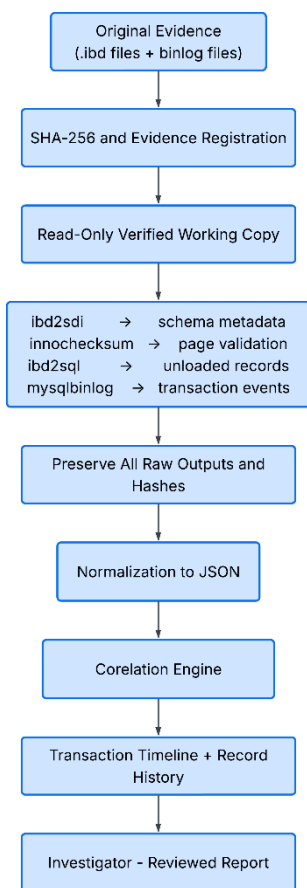


Figure 1

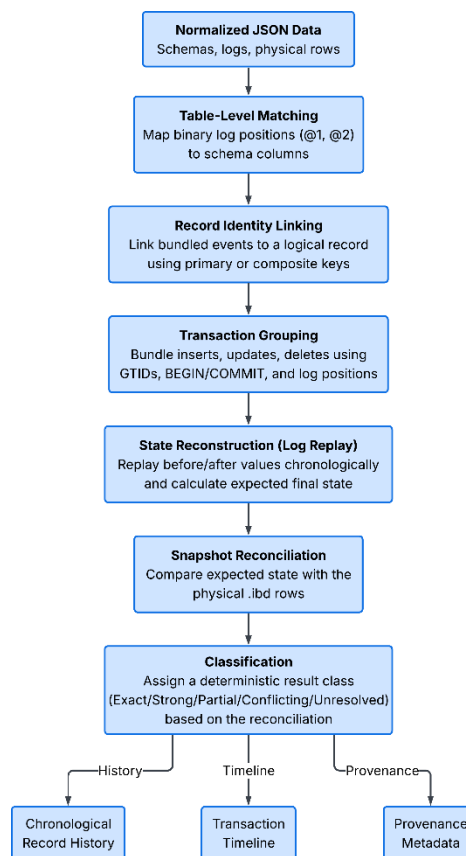


Figure 2

TECHNOLOGY STACK

To deliver a secure, heavily optimized native Linux application, we will utilize a decoupled, compiled architecture:

Framework & UI: The application will be packaged as a native Linux binary using Tauri (Rust), ensuring a secure, minimal computational footprint. The interface will be built with React to efficiently construct interactive visual elements like chronological record histories and transaction timelines.

State Management: Zustand will handle the complex global state, efficiently managing active correlation results, parsed logs, and toggling between various reconciliation views.

Backend Engine (Sidecar): A Python-based Tauri sidecar will manage the core data processing. It will programmatically orchestrate the external utilities (ibd2sql, mysqlbinlog, ibd2sdi, innochecksum), perform table-level matching, and normalize the raw outputs into the required canonical JSON model, perform correlation and stores case data in SQLite.

GUARANTEEING FORENSIC SOUNDNESS

The platform strictly enforces digital forensic principles to maintain evidence integrity and a defensible chain of custody throughout the processing pipeline:

Read-Only Integrity: The system never alters original source files. Upon intake, it immediately calculates SHA-256 hashes and generates verified working copies in a controlled workspace for all subsequent analysis.

Controlled Extraction: External extraction utilities run exclusively against the verified working copies. The tool maintains an immutable audit log detailing exact command lines, executable hashes, parser versions, outputs, and errors.

Reproducible Provenance: Every visual finding-such as reconstructed records or reconciliation conflicts-is linked directly to its raw origins. The correlation engine attaches exact provenance metadata (source file, binary-log position, transaction identifier, and raw-output reference) so examiners can independently verify all results.

IMPLEMENTATION PLAN

Implementation will proceed in four incremental phases, each producing a testable component of the pipeline described in the high-level approach.

- **Phase 1** - Build the evidence intake and integrity layer, including case registration, SHA-256 hashing, verified working-copy creation, and orchestration wrappers around ibd2sdi, innochecksum, ibd2sql, and mysqlbinlog, with full logging of commands, tool versions, and outputs.
- **Phase 2** - Implement the normalisation and correlation engine, including the canonical data model for schemas, records, log events, and transactions, and the core record-correlation and state-reconstruction logic.
- **Phase 3** - Implement timeline visualisation and reporting, producing an interactive timeline and record-history report alongside JSON, CSV, and PDF exports.
- **Phase 4** - Carry out validation, documentation, and hardening: execute the evaluation plan below, author the README and user guide, and address the failure and limitation cases identified during testing.

The investigator-facing timeline and record-history views are rendered directly in the Tauri/React desktop interface described above (Technology Stack), giving the tool a single consistent entry point rather than a separate command-line or browser-based viewer.

EVALUATION PLAN

All test data will be generated by the project team using self-created, sandboxed MySQL 8.4.x instances; no real, third-party, or production data will be used at any stage. Controlled datasets with known ground truth will include:

- Single-row insert, update, and delete operations.
- Multi-table committed transactions and a rolled-back transaction.
- Primary-key updates and a composite-key table.
- A deleted record absent from the current physical state.
- A deliberately missing binary-log file and a deliberately corrupted InnoDB page.
- Mismatched .ibd and binary-log snapshot timing, to exercise the conflict-detection path.
- Two or more concurrent sessions with interleaved committed transactions, to verify that events are correctly separated by thread ID and GTID rather than assumed to be sequential.

Because the ground truth for each dataset is known in advance, the tool's accuracy will be measured directly against expected outcomes, including transaction-boundary accuracy, record-correlation accuracy, reconciliation accuracy, false-positive and false-negative rates, provenance accuracy, and repeatability across repeated runs on the same evidence. Every unsupported condition or known limitation encountered during evaluation will be explicitly documented rather than silently absorbed into the reported results.

RISKS, LIMITATIONS, ETHICS AND LEGALITY

KEY RISKS AND LIMITATIONS

- Missing, purged, or disabled binary logs will produce a partial timeline with an explicitly reported coverage gap rather than a misleading complete one. Binary logs are not inherently tamper-resistant: an account holding the BINLOG_ADMIN privilege can remove logs via PURGE BINARY LOGS, an account holding the RELOAD privilege can remove the entire history via RESET BINARY LOGS AND GTIDS, and logs are also subject to automatic expiration (30 days by default).

- Redo logs would offer greater tamper-resistance, since their deletion is not exposed via a routine administrative command, but their circular, fixed-capacity design (commonly sized for around one hour of activity) and the lack of a maintained public parsing tool make them unsuitable as this project's primary evidence source. Correlating available redo-log data as a secondary corroboration source is noted as future work.
- The tool trusts that the external utilities (ibd2sdi, innochecksum, ibd2sql, mysqlbinlog) are themselves untampered, which is a documented assumption rather than something the tool can independently verify. ibd2sql's support for multiple MySQL versions is claimed by its maintainer rather than certified by Oracle/MySQL, and the team will independently validate its behaviour against the selected MySQL 8.4.x release.
- If the table schema changes during the period covered by the logs, events that predate the change cannot be safely mapped to the current schema; such events will be isolated and reported as unmapped rather than misinterpreted.
- Timeline accuracy depends on the source server's system clock; clock accuracy is assumed but not independently verified, and this is stated as an explicit limitation of any timing-based finding.
- When innochecksum reports a damaged or unreadable page, the tool will report the damage and proceed with partial extraction where possible, and will never fabricate or infer row values for unreadable pages.
- The tool does not draw automatic conclusions about intent or responsibility; findings are presented as evidence-backed observations for review and interpretation by a qualified examiner.

ETHICS AND LEGALITY

- All development and evaluation will use synthetic data generated on MySQL instances owned and controlled by the project team; no production, third-party, or real personal data will be used at any stage.
- The tool performs strictly read-only, offline analysis of already-acquired evidence files and does not attack, access, or modify any live target system.
- The tool is intended for use only on evidence that an investigator already has lawful, authorised access to, such as through seizure, warrant, or system ownership.

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